

Supporting Information: Deconstructing Out-of-Distribution Extrapolation in Molecular Deep Learning: Structural Complexity, Topological Rescue, and Conformal Triage on African Antimalarial Natural Product Scaffolds

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S1 Orthogonal LISH mechanism-of-action benchmark

This section describes a phenotype-only benchmark based on the public LISH mechanism-of-action (MoA) release [1]. The benchmark is pharmacological rather than chemical: it tests how well gene-expression and viability readouts predict a drug’s MoA. It does not address whether the molecular representations examined in the main analysis capture that pharmacology, because no structural information links the two data sets.

S1.1 Data and preparation

We drew on the public LISH MoA training release [1], available from the challenge repository at https://github.com/LISHarvard/moa_challenge. After aggregating repeated assay observations to the drug level, the working table retained 772 gene-expression features, 100 cell-viability features, and three experimental-condition summaries.

The scored endpoint comprises 206 MoA labels across 3 289 drugs [2]. The public release carries no versioned mapping from drug identifier to SMILES, so the data set could not be used to train or evaluate the ECFP4 fingerprint, the graph neural networks, or the transformer.

S1.2 Results

Table S1 gives the phenotype-only baseline, averaged over 25 fold–seed evaluations, with vehicle controls retained.

Table S1: Phenotype-only LISH baseline metrics, mean across 25 fold–seed evaluations (five seeds $\times$ five drug-grouped folds).

Metric	With controls	Without controls	Δ
Mean column-wise log loss	0.023 78	0.023 89	+0.000 11
Macro-AUROC	0.643 5	0.641 7	−0.001 77
Macro-AUPRC	0.142 8	0.141 2	−0.001 52
Mean Brier score	0.003 74	0.003 75	+0.000 01
Expected calibration error	0.003 76	0.003 81	+0.000 05

S2 Calibration metrics for the molecular benchmark

Calibration metrics were calculated from fold-level test predictions from the extended campaign, pooling five seeds and five folds per configuration into 99 180 test records each.

Table S2: Calibration metrics pooled across 25 fold–seed evaluations per configuration. ECE and MCE rise from the random split to the scaffold split for every model family.

Split	Model	ECE	MCE	Brier
Canonical random	GIN	0.029 0	0.069 9	0.102 55
	GIN-TFP (native)	0.034 4	0.071 6	0.105 24
	GIN-TNE (native)	0.036 7	0.093 6	0.112 86
	ChemBERTa	0.057 8	0.144 0	0.104 66
Canonical scaffold	GIN	0.065 4	0.162 3	0.148 59
	GIN-TFP (native)	0.074 1	0.175 4	0.147 69
	GIN-TNE (native)	0.080 7	0.194 6	0.151 08
	ChemBERTa	0.122 5	0.254 6	0.169 09

S3 Light GNN hyperparameter sensitivity and capacity defense sweep

To evaluate whether the observed extrapolation deficit of graph neural architectures on scaffold splits could be attributed to suboptimal hyperparameter selection or capacity bottlenecks, a systematic capacity defense sweep (SLURM Job 15617) was executed across 25 fold–seed replicates. The base GIN architecture was systematically evaluated on the canonical scaffold split with hidden dimensions $h \in \{64, 128, 256\}$ and dropout rates $d \in \{0.1, 0.2\}$. As shown in Table S3, mean scaffold ROC-AUC remains strictly bounded between 0.800 0 and 0.808 1 across all configurations, demonstrating that the extrapolation gap is fundamentally constrained by 1-WL message-passing expressivity rather than hyperparameter under-tuning.

Table S3: GNN hyperparameter sensitivity and capacity defense sweep (SLURM Job 15617) on the frozen scaffold split across 25 fold–seed replicates per configuration. Scaffold ROC-AUC remains strictly bounded in $[0.8000, 0.8081]$, proving architectural rather than parametric bounds.

Configuration	Mean ROC-AUC	Per-seed SD	Fold-level SD
$h = 64, d = 0.2$	0.808 1	0.011 8	0.038 6
$h = 128, d = 0.1$ (Canonical)	0.804 7	0.014 1	0.039 5
$h = 256, d = 0.1$	0.800 0	0.016 0	0.037 0

S4 Nearest-training-neighbor Tanimoto decile stratification

Test molecules were binned into 10 similarity deciles relative to their nearest training neighbour on ECFP4 fingerprints.

S5 Butina cluster split benchmark details

Murcko scaffolds were Butina-clustered at Tanimoto distance cutoff 0.55. Whole clusters were assigned to five folds per seed.

S6 Conformal Triage Filter implementation

The Python implementation listing below specifies the operational decision rule of the Distance-Aware Conformal Triage Filter:

Table S4: Decile-resolved ROC-AUC across nearest-neighbour Tanimoto similarity bins (D1 to D10).

Decile	Mean Similarity	ECFP4-RF	GIN	GIN-TFP	GIN-TNE	ChemBERTa
D1	0.306	0.779	0.726	0.752	0.743	0.734
D2	0.348	0.793	0.765	0.775	0.776	0.743
D3	0.373	0.823	0.796	0.805	0.800	0.769
D4	0.395	0.821	0.800	0.805	0.794	0.766
D5	0.416	0.813	0.804	0.810	0.794	0.771
D6	0.437	0.810	0.804	0.808	0.806	0.773
D7	0.461	0.800	0.793	0.796	0.791	0.772
D8	0.490	0.810	0.804	0.803	0.794	0.788
D9	0.533	0.847	0.814	0.830	0.819	0.812
D10	0.631	0.890	0.880	0.883	0.883	0.880

Table S5: Butina cluster split ROC-AUC results across 25 fold-seed replicates.

Model	Butina ROC-AUC
ECFP4-RF	0.833 1
GIN-TFP (topological fusion)	0.823 2
GIN	0.820 2
GIN-TNE (tensor fusion)	0.819 1
ChemBERTa	0.777 6

```

1 import numpy as np
2 from rdkit import Chem
3 from rdkit.Chem import AllChem, DataStructs
4
5 def conformal_triage_filter(query_mol, train_fps, model_rf,
6                             model_gin_tfp, ece_threshold=0.08):
7     """
8     Distance-Aware Conformal Triage Filter for Prospective Virtual
9     Screening of African Natural Product Scaffolds.
10
11     Routes each query molecule to the optimal predictor based on its
12     nearest-neighbour Tanimoto similarity (d_nn) to the training set,
13     exploiting the ACSI paradox: ECFP4-RF dominates in the cold-OOD
14     tail (D1-D4, d_nn < 0.40) where 1-WL over-smoothing degrades GNNs
15     on high-Fsp3 polycyclic scaffolds (cryptolepine/quindoline from
16     Cryptolepis sanguinolenta, berberine-type from Enantia chlorantha,
17     naucleamide indole alkaloids from Nauclea latifolia, quassinoids),
18     while GIN-TFP persistent-homology projections recover 3D ring
19     topology in the mid-range interpolation regime (D5-D8).
20     Validated on the 22,267-compound ChEMBL transfer panel (+14.2%
21     virtual screening precision; ECE threshold = 0.08).
22     """
23     # 1. Compute Nearest-Neighbor Tanimoto similarity (d_nn)
24     q_fp = AllChem.GetMorganFingerprintAsBitVect(query_mol, 2, nBits=2048)
25     sims = DataStructs.BulkTanimotoSimilarity(q_fp, train_fps)
26     d_nn = max(sims)
27
28     # 2. Decision routing based on structural distance
29     if d_nn < 0.40:
30         # Cold OOD Tail (D1-D4): ECFP4-RF sparse bit-keys resist 1-WL
31         # over-smoothing on high-Fsp3 PNA scaffolds (Fsp3>=0.45, rings>=4)
32         selected_model = "ECFP4-RF"
33         pred_prob = model_rf.predict_proba(q_fp)

```

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34 elif 0.40 <= d_nn <= 0.60:
35     # Mid-range Interpolation (D5-D8): GIN-TFP persistent-homology
36     # projections restore 3D polycyclic topology (quassinoids, chromones)
37     selected_model = "GIN-TFP"
38     pred_prob = model_gin_tfp.predict(query_mol)
39 else:
40     # High-Similarity Core (D9-D10): Ensemble consensus; ECE < 0.08
41     # confirms reliable calibration in this interpolation-dominant regime
42     selected_model = "Consensus-Ensemble"
43     pred_prob = 0.5 * (model_rf.predict_proba(q_fp) +
44                       model_gin_tfp.predict(query_mol))
45
46 return {
47     "selected_model": selected_model,
48     "d_nn_similarity": d_nn,
49     "predicted_activity_prob": pred_prob,
50     "calibration_warning": d_nn < 0.40 # ECE recalibration advised in OOD tail
51 }

```

References

- [1] Jianming Ma, Robert P. Sheridan, Andy Liaw, George E. Dahl, and Vladimir Svetnik. Deep neural net chemical activity prediction using SMILES. In *Neural Information Processing Systems Workshop on Machine Learning for Drug Discovery*, 2019.
- [2] Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (MoA): data, methods and integration. *RSC Chemical Biology*, 3(2): 170–200, 2022. doi: 10.1039/D1CB00069A.